

APPENDIX I

NONCONVEX OPTIMIZATION ANALYSIS OF FCPC

A. Scope of the Revised Analysis

This appendix replaces the previous convergence proof with an analysis that matches the implemented FCPC update. We make four boundaries explicit. First, the objectives used by the deep models are allowed to be nonconvex; neither strong convexity nor a contraction in distance to an unknown optimum is assumed. Second, local differential privacy (LDP) is applied only to the label histogram and sample-count metadata used for pairing, not to stochastic gradients. Third, JSDN determines the partner reference but is not assumed to be linearly related to gradient dissimilarity. Finally, the analysis does not assign a favorable sign to the partner-reference cross term. Its sign is a property to be measured, not a consequence of a large JSDN value.

These changes separate two questions. The optimization result below shows when the actual FCPC update approaches a stationary point of the global objective. The controlled pairing ablation tests whether the particular JSDN-based choice of reference improves finite-round learning relative to random and similarity-based references.

B. Algorithmic Update and Notation

To expose the effect of the FCPC term without conflating it with optimizer state, we first write the local-SGD instantiation. The global descent lemma and Theorem 1 only require the realized server direction and therefore also apply when momentum is used, by defining that direction from the net server displacement. Lemma 1 and the explicit bound (26) describe the momentum-free local-SGD instantiation; they are not presented as an exact momentum recursion.

Let $F_i : \mathbb{R}^d \rightarrow \mathbb{R}$ be client i 's local objective and let

$$F(\mathbf{w}) = \sum_{i=1}^K p_i F_i(\mathbf{w}), \quad p_i \geq 0, \quad \sum_{i=1}^K p_i = 1 \quad (1)$$

be the full-client objective. At round t , the server samples a participating set S_t . Its normalized aggregation weights are denoted by a_i^t , where $a_i^t \geq 0$ and $\sum_{i \in S_t} a_i^t = 1$. Let $m_i^t = 1$ for a paired client and $m_i^t = 0$ for the unmatched client, if one exists. Client i starts from the current global model and performs H local steps:

$$\mathbf{w}_{i,0}^t = \mathbf{w}^t, \quad (2)$$

$$\mathbf{w}_{i,s+1}^t = \mathbf{w}_{i,s}^t - \eta_t [\mathbf{g}_i(\mathbf{w}_{i,s}^t; \xi_{i,s}^t) + 2\beta_t m_i^t (\mathbf{w}_{i,s}^t - \mathbf{r}_i^t)], \quad 0 \leq s < H. \quad (3)$$

Here $\mathbf{r}_i^t$ is the paired client's previous-round model, frozen before the current local updates begin. In the first round, the unavailable history is replaced by the common initialization. An unmatched client still trains and is aggregated, but $m_i^t = 0$ removes its paired penalty in that round.

The server aggregation is

$$\mathbf{w}^{t+1} = \sum_{i \in S_t} a_i^t \mathbf{w}_{i,H}^t. \quad (4)$$

Define the effective round step size $\gamma_t = H\eta_t$ and the averaged direction

$$\mathbf{U}_t = \frac{1}{H} \sum_{s=0}^{H-1} \sum_{i \in S_t} a_i^t [\mathbf{g}_i(\mathbf{w}_{i,s}^t; \xi_{i,s}^t) + 2\beta_t m_i^t (\mathbf{w}_{i,s}^t - \mathbf{r}_i^t)]. \quad (5)$$

For this local-SGD instantiation, the server update has the exact form

$$\mathbf{w}^{t+1} = \mathbf{w}^t - \gamma_t \mathbf{U}_t. \quad (6)$$

More generally, including momentum or another base optimizer, we define $\mathbf{U}_t = (\mathbf{w}^t - \mathbf{w}^{t+1})/\gamma_t$; all results from Lemma 2 onward remain unchanged.

Let $\mathcal{F}_t$ contain all randomness revealed before round t , including $\mathbf{w}^t$ and the frozen references, and write $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_t]$. The mean direction error and conditional variance of the complete implemented round are

$$\mathbf{b}_t = \mathbb{E}_t[\mathbf{U}_t] - \nabla F(\mathbf{w}^t), \quad (7)$$

$$v_t = \mathbb{E}_t \|\mathbf{U}_t - \mathbb{E}_t[\mathbf{U}_t]\|_2^2. \quad (8)$$

The vector $\mathbf{b}_t$ includes local drift, any bias caused by client sampling and normalized aggregation, and the paired regularizer. Defining it for the complete round avoids silently treating those effects as zero-mean noise.

C. Standard Assumptions

Assumption 1 (Smooth nonconvex objective): Every F_i is differentiable and L -smooth, and the global objective is bounded below by $F_{\inf} > -\infty$.

Assumption 2 (Stochastic-gradient oracle): For every client and local iterate,

$$\mathbb{E}[\mathbf{g}_i(\mathbf{w}; \xi) \mid \mathbf{w}] = \nabla F_i(\mathbf{w}), \quad (9)$$

$$\mathbb{E}\|\mathbf{g}_i(\mathbf{w}; \xi) - \nabla F_i(\mathbf{w})\|_2^2 \leq \sigma^2. \quad (10)$$

The theorem below uses the conditional round variance v_t , which also captures client-sampling randomness and the dependence generated by multiple local steps.

No assumption connects JSDN to $\|\nabla F_i - \nabla F_j\|$. Such a bridge would require additional restrictions on the joint data distributions, for example pure label shift with shared class-conditional distributions, which are not imposed on the main experiments.

D. What the Quadratic Penalty Provably Does

Let $\mathbf{d}_{i,s}^t = \mathbf{w}_{i,s}^t - \mathbf{w}^t$. Subtracting $\mathbf{w}^t$ from both sides of (3) gives the following exact recursion.

Lemma 1 (Pullback and partner-offset decomposition): For every paired local step,

$$\mathbf{d}_{i,s+1}^t = (1 - 2\eta_t\beta_t)\mathbf{d}_{i,s}^t - \eta_t\mathbf{g}_i(\mathbf{w}_{i,s}^t; \xi_{i,s}^t) - 2\eta_t\beta_t(\mathbf{w}^t - \mathbf{r}_i^t). \quad (11)$$

If $0 \leq 2\eta_t\beta_t \leq 1$, then

$$\|\mathbf{d}_{i,s+1}^t\|_2 \leq (1 - 2\eta_t\beta_t)\|\mathbf{d}_{i,s}^t\|_2 + \eta_t\|\mathbf{g}_i(\mathbf{w}_{i,s}^t; \xi_{i,s}^t)\|_2 + 2\eta_t\beta_t\|\mathbf{w}^t - \mathbf{r}_i^t\|_2. \quad (12)$$

Proof: Use the identity

$$\mathbf{w}_{i,s}^t - \mathbf{r}_i^t = (\mathbf{w}_{i,s}^t - \mathbf{w}^t) + (\mathbf{w}^t - \mathbf{r}_i^t) \quad (13)$$

inside (3). Equation (11) follows by collecting $\mathbf{d}_{i,s}^t$, and (12) follows from the triangle inequality and $1 - 2\eta_t\beta_t \geq 0$. ■

For a global anchor, $\mathbf{r}_i^t = \mathbf{w}^t$, so the final term in (11) vanishes and the coefficient $1 - 2\eta_t\beta_t$ directly contracts accumulated local drift before the next task-gradient step. For a historical partner reference, the same pullback is present, but it is accompanied by the offset $-2\eta_t\beta_t(\mathbf{w}^t - \mathbf{r}_i^t)$. The offset is a descent contribution at the beginning of the local trajectory only when

$$\langle \nabla F(\mathbf{w}^t), \mathbf{w}^t - \mathbf{r}_i^t \rangle > 0. \quad (14)$$

Neither a large JSDN nor the quadratic penalty alone guarantees (14). Thus Lemma 1 proves parameter pullback, not unconditional gradient alignment or unconditional acceleration.

E. Single-Round Descent

Lemma 2 (Conditional one-round descent): Under Assumption 1, the update (6) satisfies

$$\begin{aligned} \mathbb{E}_t F(\mathbf{w}^{t+1}) &\leq F(\mathbf{w}^t) - \gamma_t \|\nabla F(\mathbf{w}^t)\|_2^2 - \gamma_t \langle \nabla F(\mathbf{w}^t), \mathbf{b}_t \rangle \\ &\quad + \frac{L\gamma_t^2}{2} (\|\nabla F(\mathbf{w}^t) + \mathbf{b}_t\|_2^2 + v_t). \end{aligned} \quad (15)$$

If $0 < \gamma_t \leq 1/(4L)$, then

$$\mathbb{E}_t F(\mathbf{w}^{t+1}) \leq F(\mathbf{w}^t) - \frac{\gamma_t}{4} \|\nabla F(\mathbf{w}^t)\|_2^2 + \frac{3\gamma_t}{4} \|\mathbf{b}_t\|_2^2 + \frac{L\gamma_t^2}{2} v_t. \quad (16)$$

Proof: By L -smoothness and (6),

$$F(\mathbf{w}^{t+1}) \leq F(\mathbf{w}^t) - \gamma_t \langle \nabla F(\mathbf{w}^t), \mathbf{U}_t \rangle + \frac{L\gamma_t^2}{2} \|\mathbf{U}_t\|_2^2. \quad (17)$$

Taking conditional expectation, using $\mathbb{E}_t \mathbf{U}_t = \nabla F(\mathbf{w}^t) + \mathbf{b}_t$, and applying the bias-variance identity gives (15). Next use

$$-\langle \nabla F, \mathbf{b}_t \rangle \leq \frac{1}{2} \|\nabla F\|_2^2 + \frac{1}{2} \|\mathbf{b}_t\|_2^2, \quad (18)$$

$$\|\nabla F + \mathbf{b}_t\|_2^2 \leq 2\|\nabla F\|_2^2 + 2\|\mathbf{b}_t\|_2^2. \quad (19)$$

The coefficient of $\|\nabla F\|^2$ is at most $-\gamma_t/2 + L\gamma_t^2 \leq -\gamma_t/4$, while the coefficient of $\|\mathbf{b}_t\|^2$ is at most $\gamma_t/2 + L\gamma_t^2 \leq 3\gamma_t/4$. This proves (16). ■

Equation (15) retains the cross term that was omitted in the previous proof. It shows why comparing FCPC with FedAvg cannot be reduced to comparing loss values or the magnitude of JSDN: the sign and magnitude of $\langle \nabla F, \mathbf{b}_t \rangle$, as well as the second-order cost and round variance, must all be controlled.

F. Finite-Round Nonconvex Convergence

Theorem 1 (Finite-round stationarity bound): Suppose Assumption 1 holds, $0 < \gamma_t \leq 1/(4L)$, and the quantities in (7)–(8) are finite. For every $T \geq 1$,

$$\frac{\sum_{t=0}^{T-1} \gamma_t \mathbb{E} \|\nabla F(\mathbf{w}^t)\|_2^2}{\sum_{t=0}^{T-1} \gamma_t} \leq \frac{4[F(\mathbf{w}^0) - F_{\inf}]}{\sum_{t=0}^{T-1} \gamma_t} + \frac{3 \sum_{t=0}^{T-1} \gamma_t \mathbb{E} \|\mathbf{b}_t\|_2^2}{\sum_{t=0}^{T-1} \gamma_t} + \frac{2L \sum_{t=0}^{T-1} \gamma_t^2 \mathbb{E} v_t}{\sum_{t=0}^{T-1} \gamma_t}. \quad (20)$$

Proof: Take total expectation in (16), sum it from $t = 0$ to $T-1$, and telescope the objective values. Use $\mathbb{E}F(\mathbf{w}^T) \geq F_{\inf}$ and divide by $\frac{1}{4} \sum_{t=0}^{T-1} \gamma_t$. ■

For a constant $\gamma_t = \gamma$, if $\mathbb{E} \|\mathbf{b}_t\|_2^2 \leq B^2$ and $\mathbb{E} v_t \leq V$, then

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F(\mathbf{w}^t)\|_2^2 \leq \frac{4[F(\mathbf{w}^0) - F_{\inf}]}{\gamma T} + 3B^2 + 2L\gamma V. \quad (21)$$

Thus a fixed nonvanishing direction bias generally gives convergence only to a stationary neighborhood. This is relevant to a fixed partner penalty: a historical reference that remains separated from the current global model cannot be omitted from the limiting error.

Corollary 1 (Convergence to stationarity): If $\sum_t \gamma_t = \infty$, $\sum_t \gamma_t^2 < \infty$, $\sup_t \mathbb{E} v_t < \infty$, and

$$\frac{\sum_{t=0}^{T-1} \gamma_t \mathbb{E} \|\mathbf{b}_t\|_2^2}{\sum_{t=0}^{T-1} \gamma_t} \rightarrow 0, \quad (22)$$

then the weighted average of $\mathbb{E} \|\nabla F(\mathbf{w}^t)\|_2^2$ converges to zero.

The corollary can be satisfied by a diminishing β_t together with vanishing local drift and sampling bias, or by references whose displacement from the current global model vanishes sufficiently quickly. The theorem does not claim exact stationarity for a fixed β and a persistently stale partner.

G. Where the FCPC Reference Enters the Bound

The following quantities make the role of the reference explicit without postulating a JSDN–gradient law:

$$D_t^2 = \mathbb{E}_t \left[\frac{1}{H} \sum_{s=0}^{H-1} \sum_{i \in S_t} a_i^t \|\mathbf{w}_{i,s}^t - \mathbf{w}^t\|_2^2 \right], \quad (23)$$

$$A_t^2 = \mathbb{E}_t \left[\sum_{i \in S_t} a_i^t m_i^t \|\mathbf{w}^t - \mathbf{r}_i^t\|_2^2 \right], \quad (24)$$

$$\chi_t = \mathbb{E}_t \left[\sum_{i \in S_t} a_i^t \nabla F_i(\mathbf{w}^t) \right] - \nabla F(\mathbf{w}^t). \quad (25)$$

D_t is local drift, A_t is reference staleness, and χ_t records any mismatch caused by the actual sampling and aggregation rule. By L -smoothness and Jensen's inequality,

$$\|\mathbf{b}_t\|_2^2 \leq 3L^2 D_t^2 + 3\|\chi_t\|_2^2 + 24\beta_t^2 (D_t^2 + A_t^2). \quad (26)$$

Indeed, the task-gradient drift has squared norm at most $L^2 D_t^2$, while the regularizer contribution has squared norm at most

$$4\beta_t^2 \mathbb{E}_t \left[\frac{1}{H} \sum_{s=0}^{H-1} \sum_{i \in S_t} a_i^t m_i^t \|\mathbf{w}_{i,s}^t - \mathbf{r}_i^t\|_2^2 \right] \quad (27)$$

$$\leq 8\beta_t^2 (D_t^2 + A_t^2). \quad (28)$$

Applying $\|\mathbf{x} + \mathbf{y} + \mathbf{z}\|^2 \leq 3(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + \|\mathbf{z}\|^2)$ yields (26).

Corollary 2 (Sufficient conditions for FCPC stationarity): Suppose the conditions of Theorem 1 hold, $\sum_t \gamma_t = \infty$, $\sum_t \gamma_t^2 < \infty$, and $\sup_t \mathbb{E} v_t < \infty$. If

$$\frac{\sum_{t=0}^{T-1} \gamma_t \mathbb{E} [(L^2 + 8\beta_t^2) D_t^2 + \|\chi_t\|_2^2 + 8\beta_t^2 A_t^2]}{\sum_{t=0}^{T-1} \gamma_t} \rightarrow 0, \quad (29)$$

then

$$\frac{\sum_{t=0}^{T-1} \gamma_t \mathbb{E} \|\nabla F(\mathbf{w}^t)\|_2^2}{\sum_{t=0}^{T-1} \gamma_t} \rightarrow 0. \quad (30)$$

This follows directly from (26) and the stationarity corollary above. Condition (29) exposes rather than hides the required FL controls: local drift must vanish on average, the sampling/aggregation rule must become unbiased on average, and the regularization-weighted reference staleness must vanish. For example, the last requirement can hold when the references approach the global model or when β_t decreases sufficiently quickly. A fixed β with a persistently stale partner is not covered by an exact-stationarity claim.

For a global anchor, $A_t = 0$ and Lemma 1 explains how the penalty can reduce D_t . For a historical partner, the same stabilizing pullback remains, but A_t is generally nonzero and the signed condition (14) is not guaranteed. This decomposition is consistent with the single-seed 50-round mechanism screening: the global-anchor treatment provided the clearest reduction in update dispersion and the largest observed AUC gain, whereas the high-JSDN, random, and low-JSDN partner treatments did not establish a persistent ordering in AUC at 50 rounds. The screening result motivates the decomposition; it is not used as a substitute for the proof or as a claim of statistical significance.

Consequently, JSDN does not appear as a favorable term in (20). A JSDN-based reference can improve the bound only indirectly, by empirically reducing the realized drift, reference error, sampling mismatch, or variance, or by producing a favorable cross term in (15). These mechanisms must be evaluated directly.

APPENDIX II DEFINITION AND BASIC PROPERTIES OF JSDN

Let $\mathbf{p}_i, \mathbf{p}_j \in \Delta^{C-1}$ be label distributions and let $N_i, N_j > 0$ be sample counts. With base-two logarithms, define

$$\text{JSD}(\mathbf{p}_i, \mathbf{p}_j) = \frac{1}{2} D_{\text{KL}}(\mathbf{p}_i \| \mathbf{m}_{ij}) + \frac{1}{2} D_{\text{KL}}(\mathbf{p}_j \| \mathbf{m}_{ij}), \quad (31)$$

$$\mathbf{m}_{ij} = \frac{\mathbf{p}_i + \mathbf{p}_j}{2}, \quad (32)$$

$$q_{ij} = \frac{|N_i - N_j|}{N_i + N_j}, \quad (33)$$

$$\text{JSDN}_{ij} = (1 - \lambda) \text{JSD}(\mathbf{p}_i, \mathbf{p}_j) + \lambda q_{ij}, \quad \lambda \in [0, 1]. \quad (34)$$

Proposition 1 (Boundedness, symmetry, and single-skew reduction): For every valid pair, $\text{JSDN}_{ij} = \text{JSDN}_{ji}$ and $0 \leq \text{JSDN}_{ij} \leq 1$. If $0 < \lambda < 1$, then $\text{JSDN}_{ij} = 0$ if and only if the clients have identical label distributions and equal sample counts. Under equal sample counts, JSDN and JSD induce the same pair ranking; under equal label distributions, JSDN and q_{ij} induce the same pair ranking.

Proof: Base-two JSD and q_{ij} are symmetric and belong to $[0, 1]$, so their convex combination has the same properties. For $0 < \lambda < 1$, the convex combination is zero exactly when both nonnegative components are zero. At either single-skew limit, the remaining nonzero component is multiplied by a positive constant, which preserves its ranking. ■

These properties show that JSDN is a normalized composite metadata measure; they do not prove that a larger JSDN yields a more useful optimization direction. The weight $\lambda = 0.3$ is therefore reported as an empirical choice over the tested search range, not as a universal theoretical optimum.

APPENDIX III PAIRING RULE AND ODD-CLIENT HANDLING

FCPC greedily selects a currently available edge with a large JSDN score and removes its two endpoints until fewer than two clients remain. We describe this rule as a dissimilarity-seeking heuristic. We do not use a maximum-weight-matching approximation ratio as evidence of learning quality: maximizing the sum of metadata dissimilarities is not itself a convergence or accuracy objective. Its optimization value is assessed against random and similarity-based pairing under common partitions, client samples, and seeds.

When the number of participating clients is odd, the unmatched client is not discarded. It performs the ordinary local task update, participates in server aggregation, and simply receives no paired penalty in that round. A fair-bye counter rotates this status among clients so that an always-participating client is not systematically excluded from regularization.

APPENDIX IV COMPLEXITY AND COMMUNICATION ACCOUNTING

For $n = |S_t|$ participating clients, C classes, and a model state containing P_b bytes, constructing the dense JSDN matrix costs $O(n^2 C)$ arithmetic and $O(n^2)$ storage. Sorting all candidate edges and scanning them greedily costs $O(n^2 \log n)$ time and $O(n^2)$ storage. Computing the quadratic penalty adds $O(P)$ arithmetic per mini-batch over P trainable parameters and requires one frozen reference state.

FCPC does not add server communication rounds, but a full-model partner implementation adds partner-model traffic. With an even participating set, the model payload is $3nP_b$ per round for FCPC versus $2nP_b$ for FedAvg: nP_b for global downloads, nP_b for server uploads, and an additional nP_b for partner references. This 50% model-payload increase is measured and reported rather than called negligible.

APPENDIX V

THREAT MODEL

The statistical analysis assumes honest-but-curious clients and an authenticated server. Metadata LDP does not make the method robust to a malicious client that sends a poisoned partner model. Because FCPC retains the base server aggregator, it can in principle be combined with robust aggregation or anomaly detection, but both server-bound and partner-bound states must then be filtered. No Byzantine-robustness claim follows from dissimilarity-aware pairing alone.

REFERENCES

- [1] L. Yang, Y. Miao, Z. Liu, Z. Liu, X. Li, D. Kuang, H. Li, and R. H. Deng, "Enhanced model poisoning attack and multi-strategy defense in federated learning," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 3877–3892, 2025.